Experiment No: 1

Course Outcome: LO **Blooms Level: L2**

Aim: Prepare a case study on AI Agents including type, environment, and PEAS properties.

Abstract:

This experiment focuses on analyzing different AI task environments using the PEAS framework, which specifies Performance measures, Environment, Actuators, and Sensors for an intelligent agent. Students systematically describe PEAS for multiple real-world scenarios and then relate these to key task environment properties such as observability, determinism, and episodic versus sequential behavior. Through this, students gain a structured understanding of how to model and compare task environments for rational agent design.

Sample Input:

1. Agent: Part-picking robot
   Performance measure: Percentage of parts in correct bins
   Environment: Convey p, or belt with parts, bins
   Actuators: Joint arm and hand
   Sensors: Camera, joint angle sensors
2. Agent: Automated taxi driver
   Performance measure: Safe, fast, legal, comfortable trip, maximize profits
   Environment: Roads other traffic Environment: Roads, other traffic, pedestrians, customers
   Actuators: Steering wheel, accelerator, brake, signal, horn
   Sensors: Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard

Theory:

An agent is something that perceives and acts in an environment. The agent function for an agent specifies the action taken by the agent in response to any percept sequence. An agent is anything that can be viewed
as perceiving its environment through sensors and acting upon that environment through actuators.

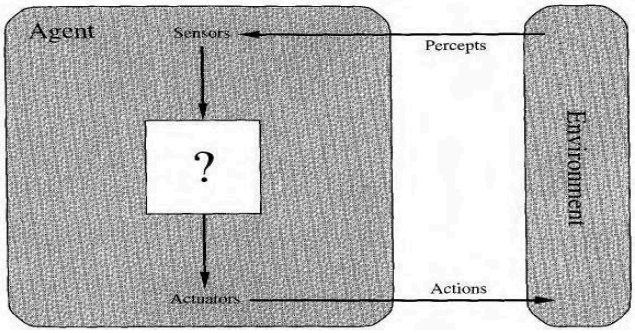


Figure: Agents interact with environments through sensors and actuators.

A performance measure embodies the criterion for success of an agent's behavior. When an agent is plunked down in an environment, it generates a sequence of actions according to the percepts it receives. This sequence of actions causes the environment to go through a sequence of states. If the sequence is desirable, then the agent has performed well.

A rational agent acts so as to maximize the expected value of the performance measure, given the percept sequence it has seen so far. A task environment specification includes the performance measure, the external environment, the actuators, and the sensors. In designing an agent, the first step must always be to specify the task environment as fully as possible. Task environments vary along several significant dimensions. They can be fully or partially observable, deterministic or stochastic, episodic or sequential, static or dynamic, discrete or continuous, and single-agent or multiagent.

The task environment for an agent is comprised of PEAS (Performance measure, Environment, Actuators, Sensors)

PEAS specify the setting of an intelligent agent:

- **P**: The **p**erformance measure defines degree of success.
- **E**: What does the agent know about the **e**nvironment?
- **A**: The **a**ctions that the agent can perform.
- **S**: Everything that an agent has perceived so far through its **s**ensors.

Output:

Example of Agent types and their PEAS (performance, environment, actuators, sensor) Descriptions.

1. English Tutor Example: PEAS
2. Robot: PEAS
3. Medical Example: PEAS

4. Taxi Example – PEAS
5. Chess player
6. Internet shopping Agent
7. Spam Filter
8. Automated taxi driver
9. Part-Sorting Robot

Conclusion:

For any given problem for a particular agent the required performance measure to measure the accuracy of an agent, the environment behavior of an agent, the sensors required to receive the input from the environment and the actuators required to perform the task is carefully observed and analyzed.

Exercise 1:

Student will choose a 3-4 familiar mobile app (e.g., navigation app, food-delivery app) and write their detailed PEAS description, then identify at least four properties of its task environment, such as fully/partially observable, static/dynamic, and discrete/continuous and prepare a comparison table.

Exercise 2:

Students will take one of their PEAS descriptions (for example, a home robot or speech assistant) and change the task environment, such as moving from home to hospital or from single user to multiuser setting. They will then update the PEAS components to reflect the new environment and write a brief note on how these changes impact agent design and complexity.